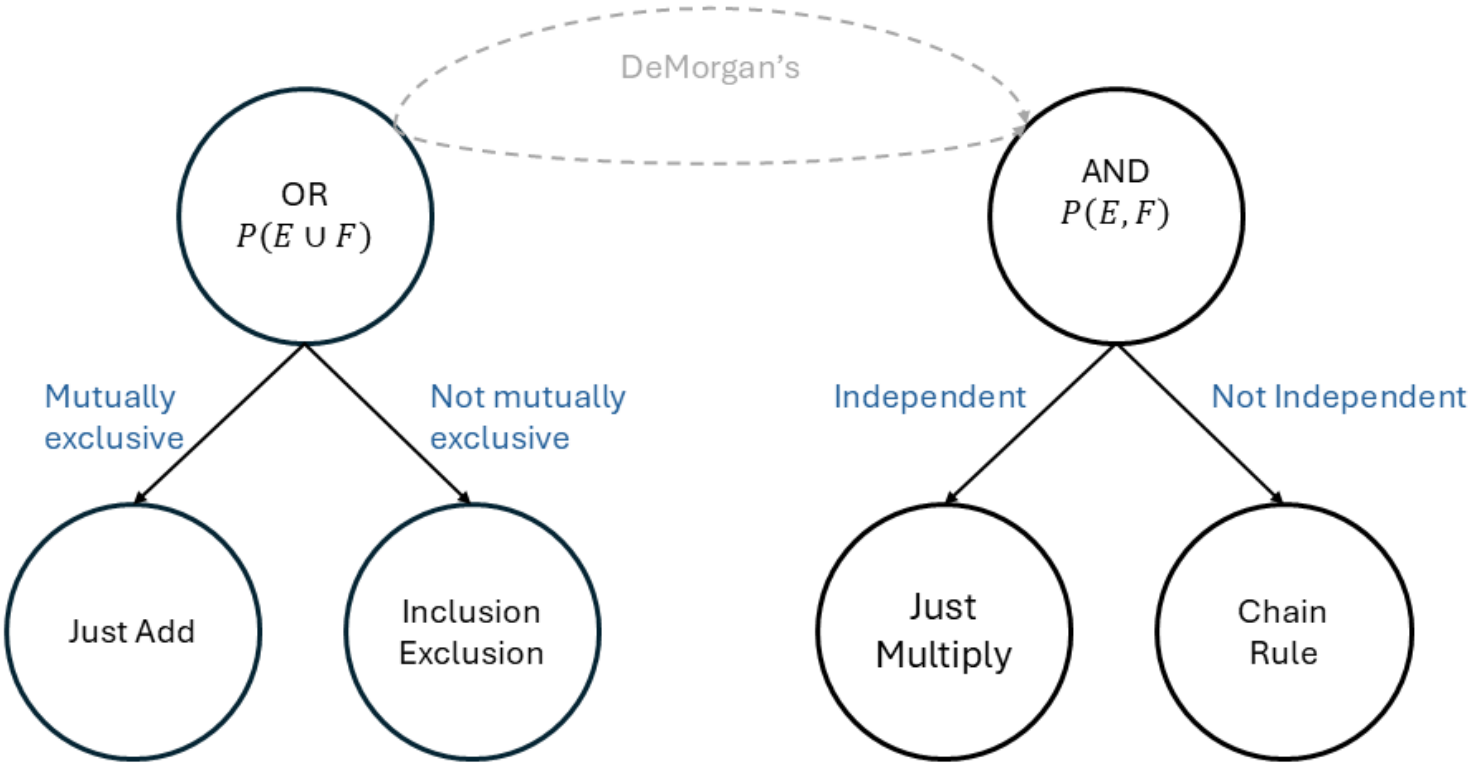
Probability and Statistic (for CS Students)

Lecture 8: Random Variables

University of Mazandaran

Fall 1405

Recap: Probability of Or and And



Recap: Conditional Probability

Two events A and B are **conditionally independent** given an event C with $P(C) > 0$ if:

$$P(A, B|C) = P(A|C)\, P(B|C)$$

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Equivalently:

$$P(A|B, C) = P(A|C)$$

The Most Important Probability Question

Say, a coin comes up heads with probability p

- Flip the coin n times
- Each coin flip is an independent trial
- What is the probability of exactly k heads?

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Finally: The probability of exactly $k = 4$ flips are heads

$$\binom{n}{k} p^k (1-p)^{n-k}$$

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- $P(Y = 3) = 1/8 \quad [H, H, H]$
- $P(Y \geq 3) = 0$

R.V.s and Events

It is confusing that both R.V.s and events use the same notation

R.V.s and Events

Consider 5 coin flips, each independently come up heads with probability p

- Recall:

$$P\left(2 \text{ heads}\right) = \binom{5}{2} p^2 \left(1-p\right)^3$$

$$P\left(3 \text{ heads}\right) = \binom{5}{3} p^3 \left(1-p\right)^2$$

- $Y =$ number of "heads" on 5 flips

$$Y \in \{0, 1, 2, \cdots, 5\}$$

$$P\left(Y = k\right) = \binom{5}{k} p^k \left(1-p\right)^{5-k}$$

Properties of R.V.s

- Probability Mass Function (PMF): $P(X = a)$
- Expectation: $E[X]$
- Variance: $\mathrm{Var}(X)$

Probability Mass Function

Probability Mass Function

The relationship between values a r.v. can take on, and the corresponding probability, is a *function*!

Probability Mass Function

Let Y be a random variable

$$Y$$

For example, Y is the number of heads in 5 coin flips

Probability Mass Function

$$Y = 2$$

It is an event when it takes on a value

For example, Y is the number of heads in 5 coin flips

Any Boolean operation applies to a R.V. can make it an event

Probability Mass Function

If this is a number



$$P(Y = 2)$$

For example, Y is the number of heads in 5 coin flips

Probability Mass Function

If this is a number

$$P(Y = 2)$$

Then, this is a number (between 0 and 1)

For example, Y is the number of heads in 5 coin flips

Probability Mass Function

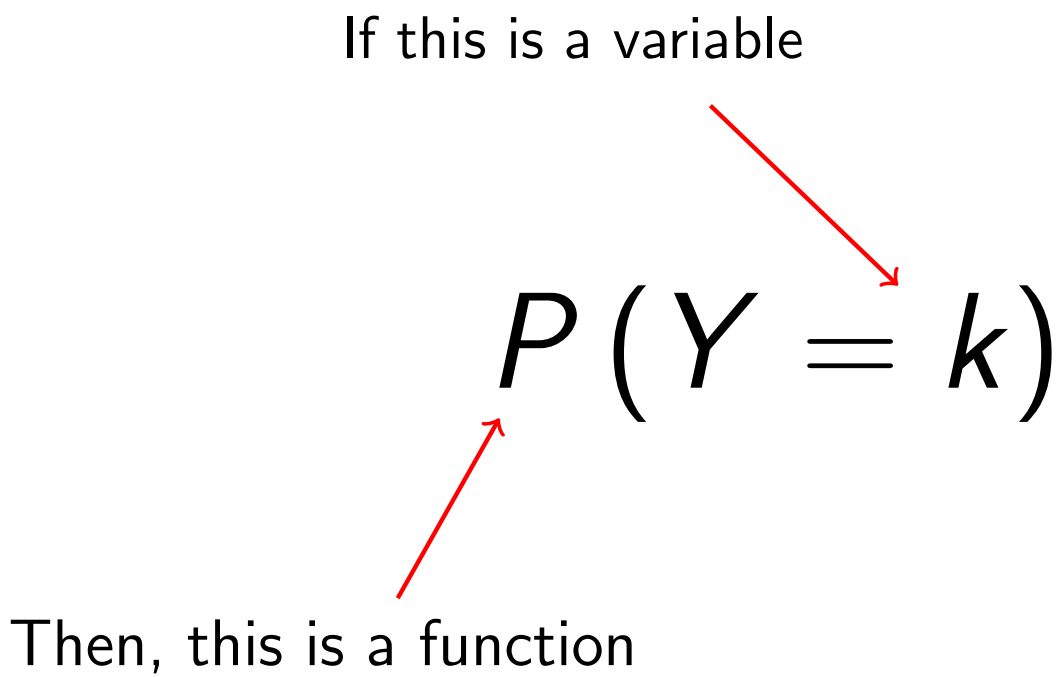
If this is a variable



$$P(Y = k)$$

For example, Y is the number of heads in 5 coin flips

Probability Mass Function



For example, Y is the number of heads in 5 coin flips

Probability Mass Function

If a random variable is discrete, we call this function

Probability Mass Function

Probability Mass Function

Let X be a random variable that represents the result of a **single die roll**. X can take on the values $\{1, 2, 3, 4, 5, 6\}$

$$P(X = x)$$

Probability Mass Function

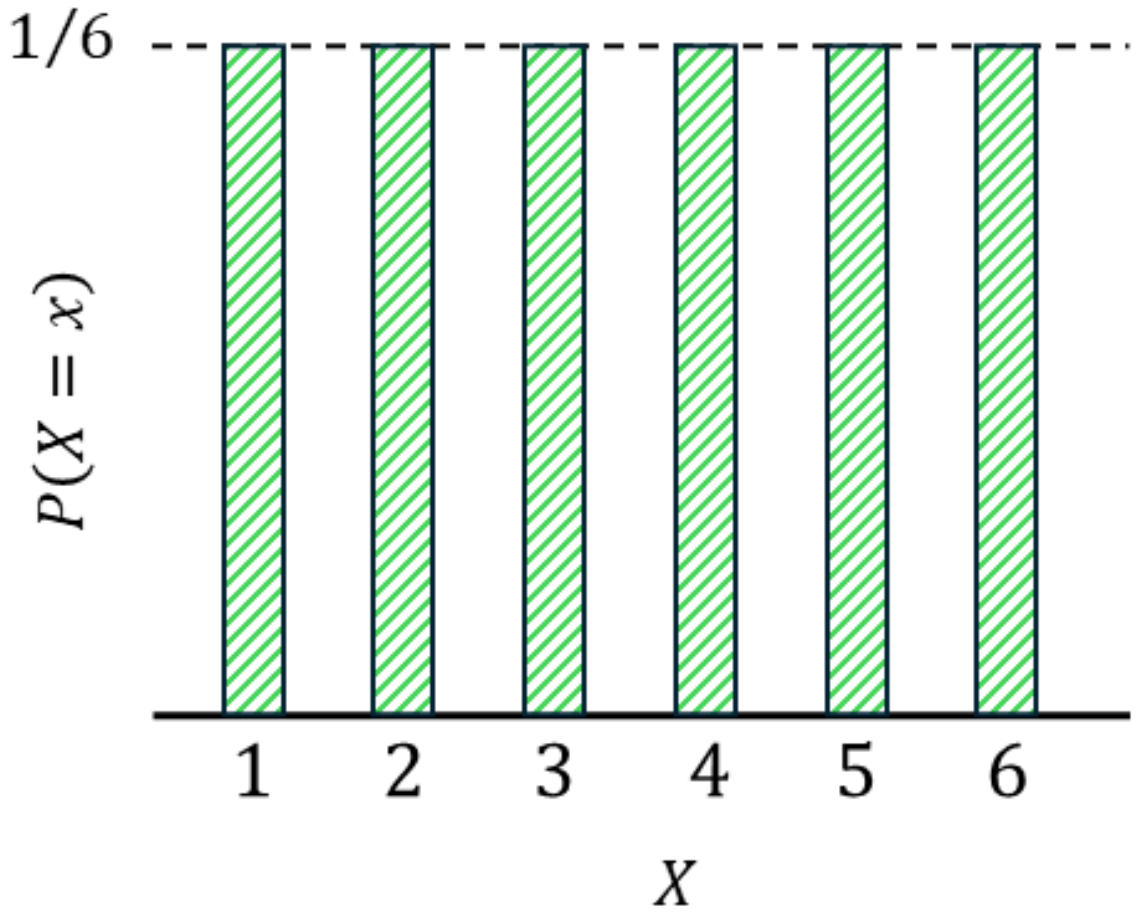
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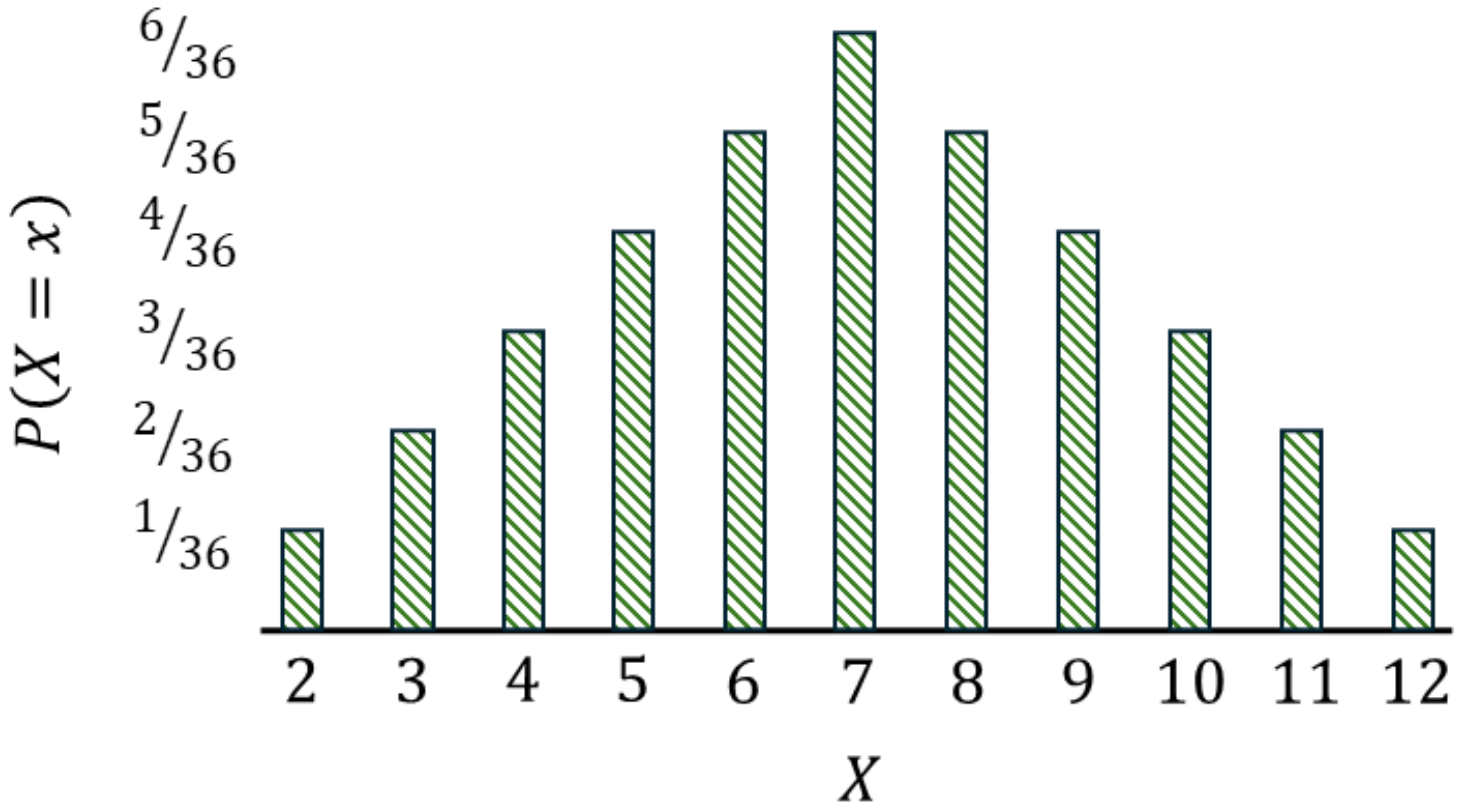
$p(x)$ a shorthand notation for the PMF

$p_x(x)$ another shorthand notation for the PMF

PMF For The Outcome of a Die Roll



PMF For The Sum of Two Dice Rolls



PMF as an Equation

$$P(X = x) = \begin{cases} \frac{x-1}{36} & \text{if } x \in \mathbb{Z}, 1 \leq x \leq 6 \\ \frac{13-x}{36} & \text{if } x \in \mathbb{Z}, 7 \leq x \leq 12 \\ 0 & \text{else} \end{cases}$$

Sum of Probabilities of a R.V.

$$\sum_{\text{all } k} P(Y = k) = ?$$

Sum of Probabilities of a R.V.

$$\sum_{\text{all } k} P(Y = k) = 1$$

Expectation

Expectation

The Expected Values for a discrete random variable X is defined as:

$$E[X] = \sum_{x:p(x)>0} x.P\left(X = x\right)$$

Note: sum over all values of x that have $p\left(x\right) > 0$

Expected value also called: **Mean**, Expectation, **Weighted Average**, **Center of Mass**, 1st Moment

Expectation from Data

$$X = [3, 2, 6, 10, 1, 1, 5, 4, \cdots]$$

$$E[X] = \sum_{x} x.P\left(X = x\right)$$

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$$\approx \frac{1}{N} \sum_{\text{values } v} v$$

Expected Value

Roll a 6-sided die. X is outcome of roll

- $P(X=1) = P(X=2) = P(X=3) = P(X=4) = P(X=5) = P(X=6) = 1/6$

$$E[X] = 1\left(\frac{1}{6}\right) + 2\left(\frac{1}{6}\right) + 3\left(\frac{1}{6}\right) + 4\left(\frac{1}{6}\right) + 5\left(\frac{1}{6}\right) + 6\left(\frac{1}{6}\right)$$

Lying With Statistics

Mark Twain: "There are three kinds of lies: lies, damned lies, and statistics"

A School has 3 classes with 5, 10, and 150 students

Randomly choose a class with equal probability

$X =$ size of chosen class

What is $E[X]$?

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- $E[X] = 5\left(\frac{1}{3}\right) + 10\left(\frac{1}{3}\right) + 150\left(\frac{1}{3}\right) = 55$

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Randomly choose a student with equal probability

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What is $E[Y]$?

- $E[Y] = 5\left(\frac{5}{165}\right) + 10\left(\frac{10}{165}\right) + 150\left(\frac{150}{165}\right) \approx 137$

Properties of Expectation

Linearity:

$$E[aX + b] = aE[X] + b$$

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$$E[aX + b] = aE[X] + b$$

$$\begin{aligned} E[aX + b] &= \sum_x (aX + b)\, p(x) \\ &= \sum_x (aXp(x) + bp(x)) \\ &= \sum_x aXp(x) + \sum_x bp(x) \\ &= a\sum_x Xp(x) + b\sum_x p(x) \\ &= aE[X] + b \end{aligned}$$

Properties of Expectation

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Expectation of a sum is the sum of expectations:

$$E[X + Y] = E[X] + E[Y]$$

Properties of Expectation

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X	Y	$X+Y$
3	4	7
2	2	4
6	8	14
10	23	33
1	-3	-2
1	0	1
5	9	14
4	1	5
$\cdots$	$\cdots$	$\cdots$
$E[X]$	$E[Y]$	$E[X+Y]$

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Unconscious Statistician:

$$E[g(X)] = \sum_x g(x)\, p(x)$$

Feedback Time!

Help us improve the course quality by giving us feedback here

